

Neutrinos Workflow Builder — Reference Guide

1. Overview

The Workflow Builder is the visual engine behind every Neutrinos Flow. This reference guide covers node types, error handling, versioning, and common troubleshooting scenarios reported by builders.

2. Node Types

- **Trigger** — starts a flow: Manual, Scheduled (cron-based), Webhook, or Event (e.g., record created in a connected database).
- **Action** — performs a task: API call, database operation, send notification, transform data.
- **Condition** — branches the flow based on a boolean expression (if/else, switch-case).
- **Loop** — iterates over a collection (e.g., process each row in a batch upload).
- **Human Task** — pauses the flow and waits for a person to approve, reject, or fill a form before continuing.
- **Sub-flow** — calls another published Flow as a reusable component.

3. Error Handling & Retries

Every Action and Connector node has a configurable retry policy: number of retries (default 3), backoff interval (default exponential, starting at 5s), and a fallback path that executes if all retries are exhausted.

Best practice: always attach a fallback path on external Connector calls (e.g., third-party APIs) to log the failure and notify an owner, rather than letting the flow silently terminate.

3.1 Common Errors

- **Timeout (Error 504)** — the external system did not respond within the configured timeout window (default 30s). Increase timeout for known slow systems or add a retry with backoff.
- **Connector Not Authorized (Error 401)** — credentials missing or expired for the current environment. Re-authenticate the connector in Environment Settings.
- **Schema Mismatch (Error 422)** — data passed to a node does not match the expected input schema. Check the Data Mapper step immediately preceding the failing node.
- **Loop Limit Exceeded** — a Loop node exceeded the default max iteration guard (1000). Confirm the source collection size or raise the guard limit in Loop Settings if intentional.

4. Versioning & Publishing

Flows are versioned automatically on every publish. Only published versions can be promoted between environments; in-progress Dev changes are always considered a draft.

Rollback: from the Version History panel, any prior published version can be restored as the active version in a given environment with one click. Rollback does not affect Dev drafts.

5. Troubleshooting: Flow Stuck at a Human Task node

This is one of the most common support questions. A Flow appears 'stuck' when it reaches a Human Task node and is simply waiting for the assigned approver to act — this is expected behavior, not a bug.

To resolve: check the Human Task's assignee configuration (a specific user, a role, or a queue). If assigned to a role with no active members, the task will wait indefinitely. Reassign via the Run Monitor's 'Reassign Task' action.

Escalation: configure an SLA timer on the Human Task node (Settings > SLA) to auto-escalate to a backup approver or send a reminder notification after a defined period.